

Sensor-Based Intelligent Accident Detection & Emergency Response System

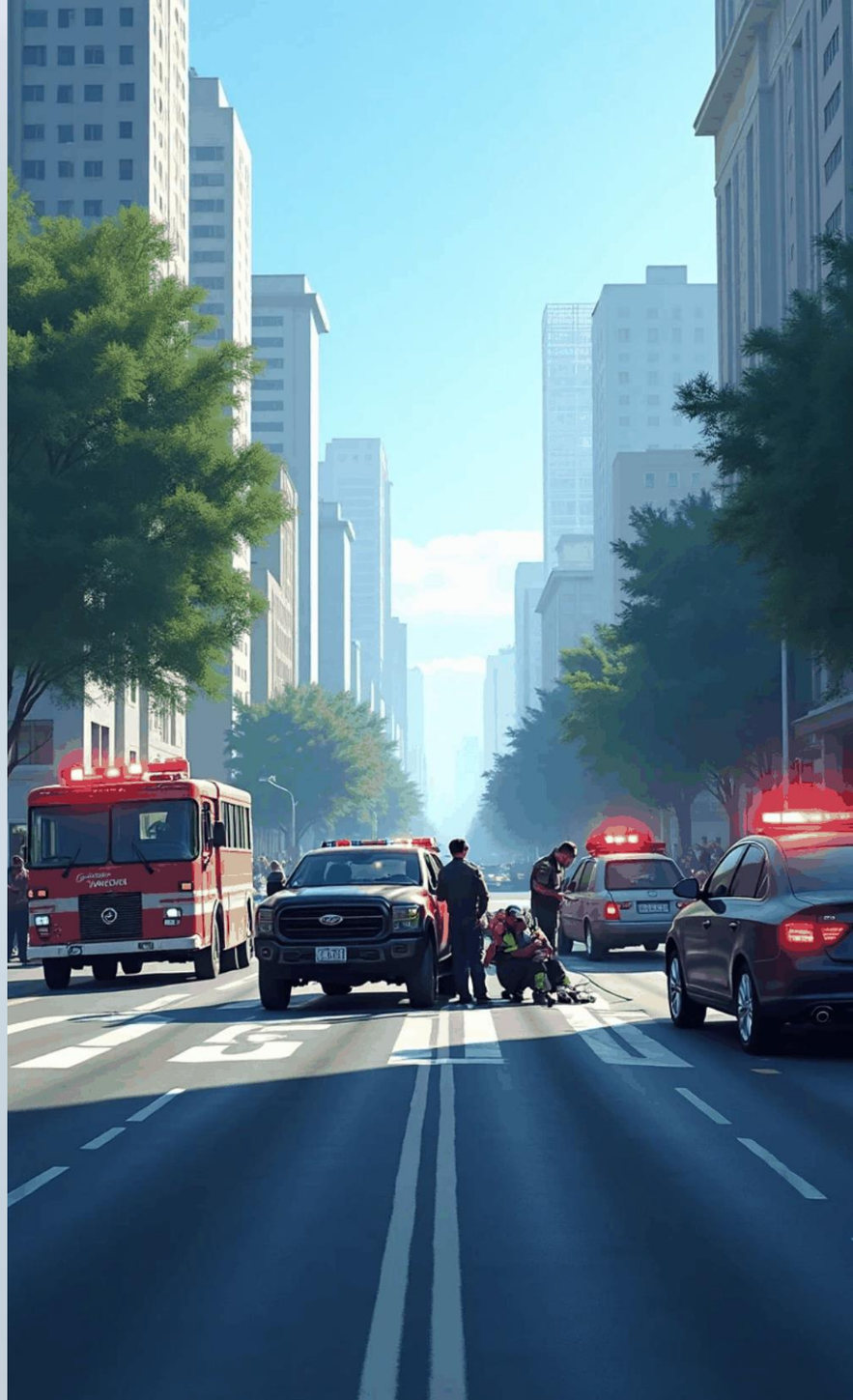
CONTENTS

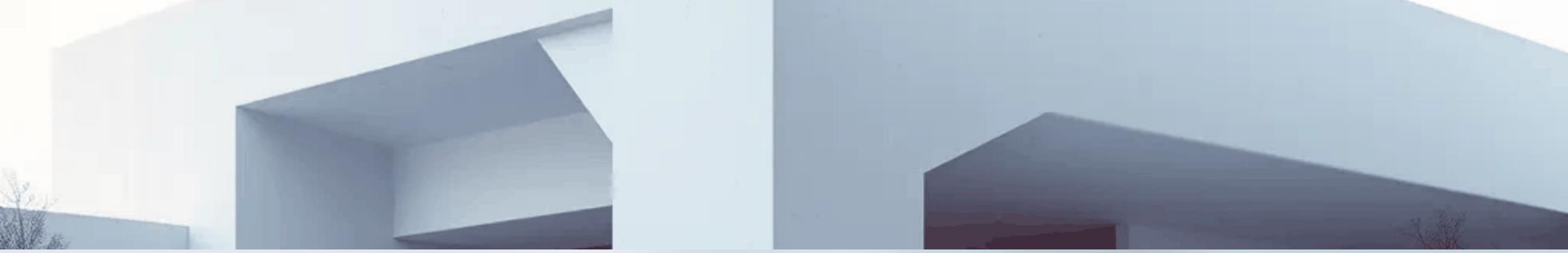
1. Introduction
2. Problem Statement
3. Impact of the Problem
4. Objectives
5. AI-Powered Solution (Overview)
6. Specific AI Tools
7. Technical Architecture
8. Data Flow
9. Live Demo (Concept Flow)
10. Key Features & Innovation
11. Expected Impact
12. Budget Plan
13. Future Scope
14. Conclusion

Introduction

India alone accounts for 11% of global accident deaths. In many cases, victims don’t receive medical care within the golden hour.

Technology can bridge the gap between accident → alert → emergency response.





Problem Statement

High Fatality Rate: Accidents are the 8th leading cause of death globally.

Response Delay: Average delay = 15–30 minutes before medical help arrives.

Manual Reporting Flaws: Witnesses may delay or fail to report.

Target Audience Impact:

Vehicle owners → increased safety.

Families → peace of mind.

Hospitals → faster response.

Governments → better road safety insights.

Impact of the Problem

Golden Hour Concept: Chances of survival drop drastically if help arrives late.

Rural/remote areas face even longer delays in ambulance arrival. Insurance claims often lack verified accident data.

Governments lack real-time accident mapping for policy-making.



Objectives

Accurate Accident Detection: Use IoT sensors (accelerometer, gyroscope).

Automated Alerts: Emergency messages sent via GSM/IoT.

AI-Based Filtering: Reduce false alarms (hard braking, potholes).

Reliable Communication: Dual channel (GSM + IoT) ensures alerts even in weak networks.

Real-Time Monitoring: Cloud dashboard for hospitals, police, and fleet operators.





AI-Powered Solution (Overview)

Sensors: Detect sudden impact, tilt, or abnormal movement.

Microcontroller + AI: Decides whether the event = real accident.

GPS: Fetches exact accident coordinates.

GSM/IoT Module: Sends SMS/WhatsApp/call alerts.

Cloud Dashboard: Stores and visualizes data for multiple vehicles.

Specific AI Tools

TensorFlow Lite / Scikit-learn – lightweight ML model for accident detection.

Google Maps API – integrates GPS for real-time location sharing.

Twilio/WhatsApp API – automatic alerts to emergency contacts.

AWS IoT Core / Google Cloud IoT – scalable monitoring and storage.

Edge AI Processing: Local inference reduces cloud dependency.





Technical Architecture

System Flow:

Input Layer

Accelerometer, gyroscope, impact sensors.

Processing Layer

ESP32/Arduino + ML model (accident detection).

Communication Layer

GSM/IoT → Sends alerts (SMS/WhatsApp).

Cloud Layer

Accident data stored + dashboard updated.

End Users: Hospitals, ambulance services, families, government dashboards.

Data Flow

Step 1

Accident occurs → Sensors record abnormal values.

Step 2

Microcontroller collects sensor data.

Step 3

AI Model confirms → accident vs false alarm.

Step 4

GPS retrieves location.

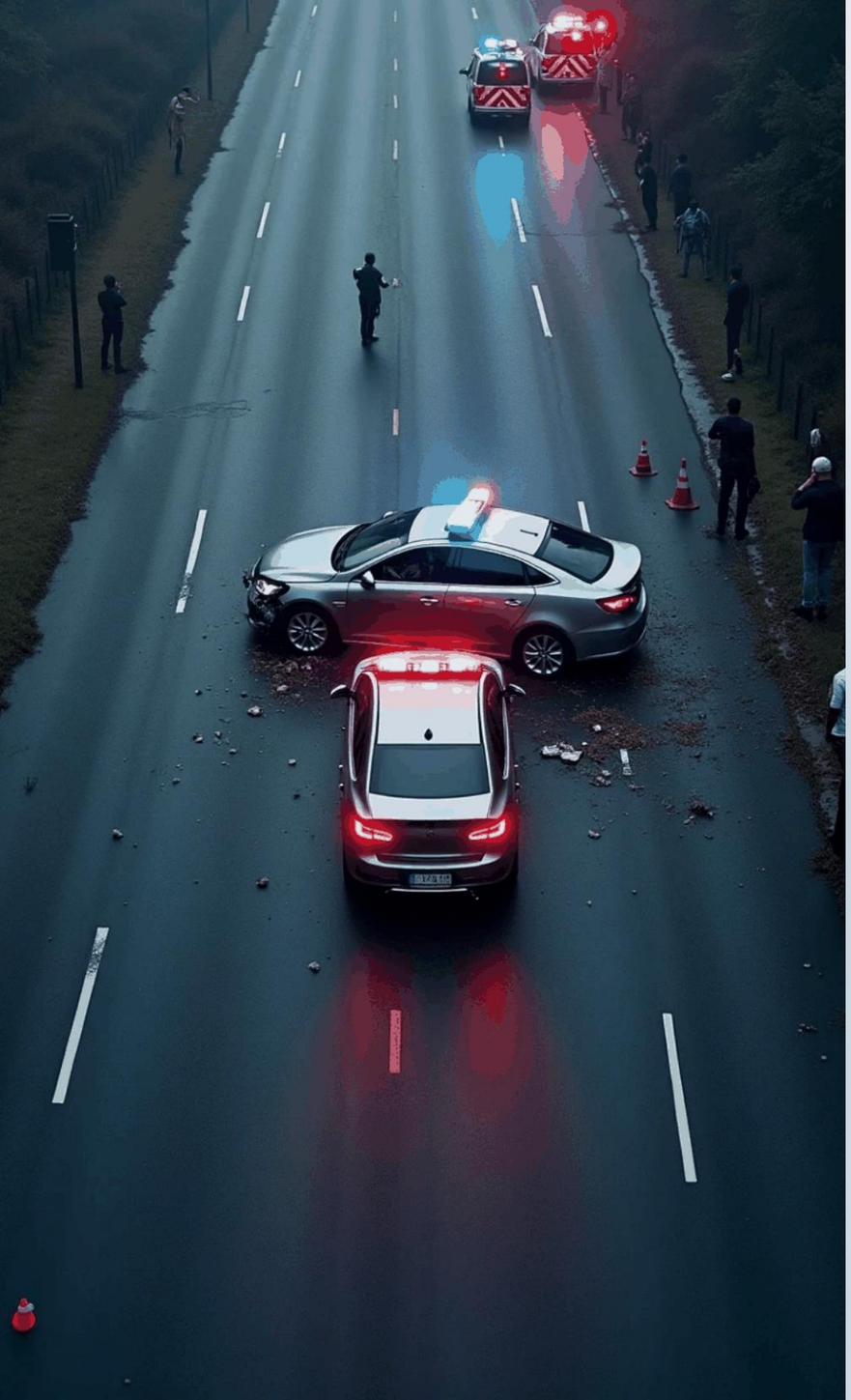
Step 5

GSM/IoT transmits alerts.

Step 6

Cloud dashboard visualizes event in real time.





Live Demo (Concept Flow)

Accident Scenario: Vehicle collides → sensors trigger.

Decision Making: AI confirms real accident.

Alert Transmission: GPS + GSM send emergency notifications.

Visualization: Accident instantly appears on cloud dashboard with location & time.

Notification Recipients: Hospital, ambulance, police, and family.



Key Features & Innovation

Real-Time Detection – instant sensor trigger.

Automatic Alerts – no human action needed.

Hybrid Communication – GSM + IoT ensures reliability.

Predictive Analytics – identify accident-prone roads/zones.

Scalable Design – adaptable for bikes, cars, buses, and fleets.



Expected Impact

Faster Medical Response: Emergency help within minutes.

Reduced Fatalities: Save more lives by acting within golden hour.

Improved Road Safety: Authorities get real-time data on accident-prone zones.

Insurance Benefits: Automated, verified accident records = faster claims.

Smart Cities: Supports government’s smart traffic & safety initiatives.

Budget Plan

Item	Qty	Unit Cost (₹)	Total (₹)
Accelerometer + Gyroscope	2	250	500
Impact / vibration sensor	1	100	100
GPS Module	1	600	600
GSM Module	1	600	600
Microcontroller (ESP32)	1	600	600
Power + Battery	1	600	600
Misc. components + SIM/recharge	—	1,100	1,100
Cloud IoT + APIs	—	0	0
Assembly and testing	—	1,500	1,500
Total			5,600

Future Scope

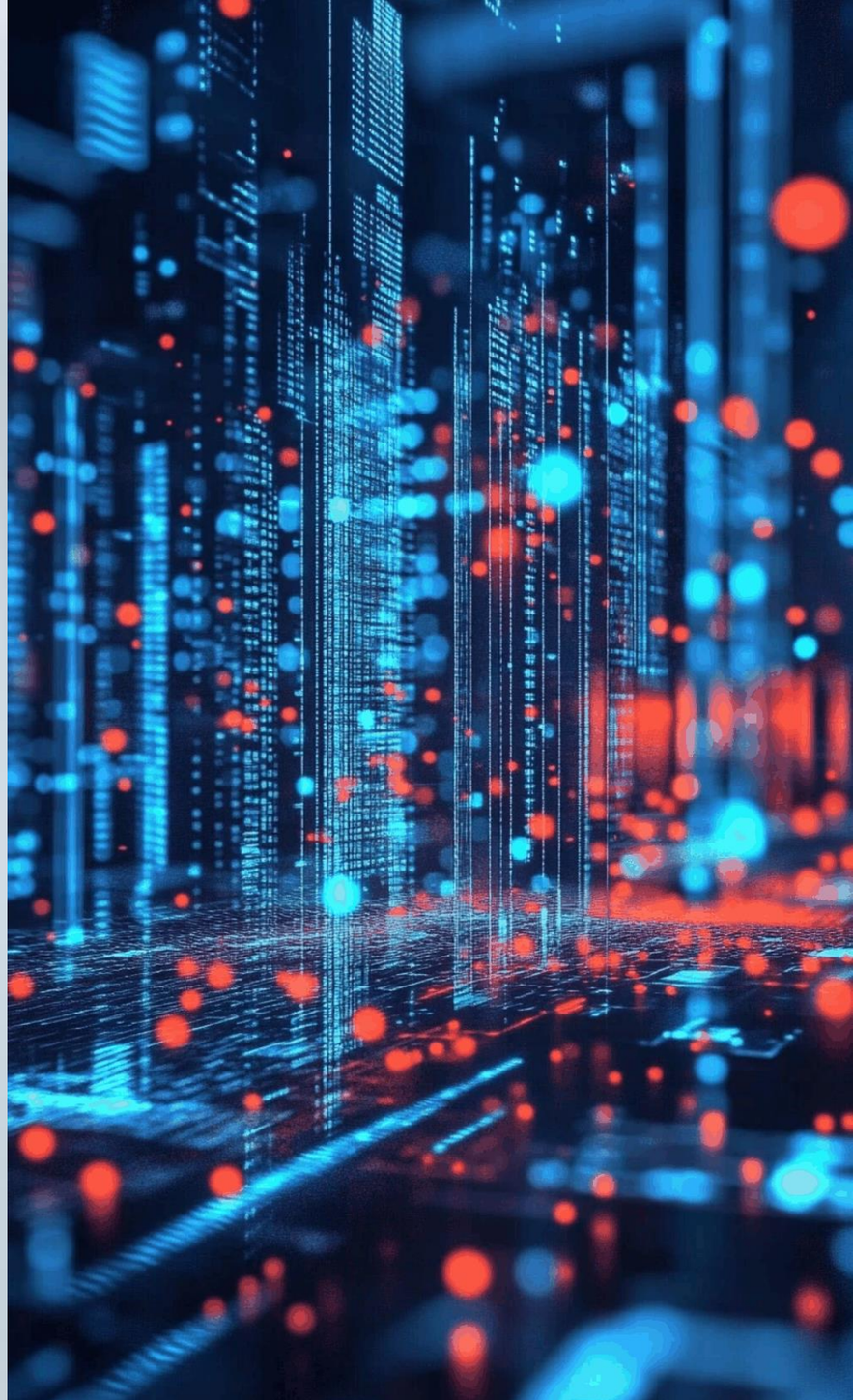
Predictive Accident Prevention: Use AI to analyze driving patterns.

Insurance Automation: Instant claims from verified accident data.

Government Dashboards: Provide real-time accident mapping.

Plug-and-Play Kits: Installable in old & new vehicles.

Smart City Integration: Connect with smart traffic lights & autonomous cars.





Conclusion

Innovative: AI + IoT for life-saving applications. Scalable: Can be deployed across vehicles & cities. Affordable: Prototype at ₹5,600; cost reduces in bulk.

Impactful: Saves lives, informs policy, and supports smart cities. Vision: A world where accidents are not the end, but the beginning of immediate help.

Thank You